KEMENTERIAN
PENDIDIKAN
MALAYSIA

KOLEJ MATRIKULASI PERAK

PRA-PSPM
SET 1

JANGAN BUKA KERTAS SOALAN INI SEHINGGA DIBERITAHU.
DO NOT OPEN THIS QUESTION PAPER UNTIL YOU ARE TOLD TO DO SO.

Name: ____________________

Matric. No.: ____________________

Class: __________

NO.	MARKS	
1	8	
2	12	
3	9	
4	12	
5	9	
6	12	
7	18	
TOTAL	**80**	

INSTRUCTION TO CANDIDATE:

This question paper consists of **7** questions. Answer **all** the questions.
The use of electronic calculator is permitted.

LIST OF SELECTED CONSTANT VALUES

Speed of light in vacuum	c	$= 3.00 \times 10^{8}$ m s^{-1}
Permeability of free space	μ_o	$= 4\pi \times 10^{-7}$ H m^{-1}
Permittivity of free space	ε_o	$= 8.85 \times 10^{-12}$ F m^{-1}
Elementary charge magnitude	e	$= 1.60 \times 10^{-19}$ C
Planck constant	h	$= 6.63 \times 10^{-34}$ J s
Electron mass	m_e	$= 9.11 \times 10^{-31}$ kg
		$= 5.49 \times 10^{-4}$ u
Neutron mass	m_n	$= 1.674 \times 10^{-27}$ kg
		$= 1.008665$ u
Proton mass	m_p	$= 1.672 \times 10^{-27}$ kg
		$= 1.007277$ u
Deuteron mass	m_d	$= 3.34 \times 10^{-27}$ kg
		$= 2.014102$ u
Molar gas constant	R	$= 8.31$ J K^{-1} mol^{-1}
Rydberg constant	R_H	$= 1.097 \times 10^{7}$ m^{-1}
Avogadro constant	N_A	$= 6.02 \times 10^{23}$ mol^{-1}
Boltzmann constant	k	$= 1.38 \times 10^{-23}$ J K^{-1}
Gravitational constant	G	$= 6.67 \times 10^{-11}$ N m^{2} kg^{-2}
Free-fall acceleration	g	$= 9.81$ m s^{-2}
Atomic mass unit	1 u	$= 1.66 \times 10^{-27}$ kg
		$= 931.5\ \dfrac{\text{M}e\text{V}}{c^2}$
Electron volt	1 eV	$= 1.6 \times 10^{-19}$ J
Constant of proportionality for Coulomb's law	$k = \dfrac{1}{4\pi\varepsilon_o}$	$= 9.0 \times 10^{9}$ N m^{2} C^{-2}
Atmospheric pressure	1 atm	$= 1.013 \times 10^{5}$ Pa
Density of water	ρ_w	$= 1000$ kg m^{-3}

LIST OF SELECTED FORMULAE
SENARAI RUMUS TERPILIH

1. $v = u + at$
2. $s = ut + \frac{1}{2}at^2$
3. $v^2 = u^2 + 2as$
4. $s = \frac{1}{2}(u + v)t$
5. $v = u - gt$
6. $v^2 = u^2 - 2gs$
7. $s = ut - \frac{1}{2}gt^2$
8. $p = mv$
9. $J = F\Delta t$
10. $\vec{J} = \Delta\vec{p} = m\vec{v} - m\vec{u}$
11. $f_s \leq \mu_s N$
12. $f_k = \mu_k N$
13. $W = \vec{F} \cdot \vec{s} = Fs\cos\theta$
14. $K = \frac{1}{2}mv^2$
15. $U = mgh$
16. $U_s = \frac{1}{2}kx^2 = \frac{1}{2}Fx$
17. $a_c = \frac{v^2}{r} = r\omega^2 = v\omega$
18. $F_c = \frac{mv^2}{r} = mr\omega^2 = mv\omega$
19. $s = r\theta$
20. $v = r\omega$
21. $a_t = r\alpha$
22. $\omega = \omega_o + \alpha t$
23. $\theta = \omega_o t + \frac{1}{2}\alpha t^2$
24. $\omega^2 = {\omega_o}^2 + 2\alpha\theta$
25. $\theta = \frac{1}{2}(\omega_o + \omega)t$
26. $\tau = rF\sin\theta$
27. $\frac{Q}{t} = -kA\left(\frac{\Delta T}{L}\right)$
28. $v_{rms} = \sqrt{< v^2 >}$
29. $v_{rms} = \sqrt{\frac{3kT}{m}} = \sqrt{\frac{3RT}{M}}$
30. $\Delta U = Q - W$

1. (a)

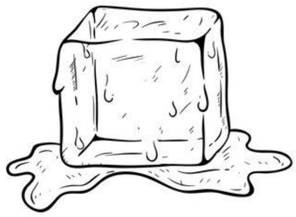

FIGURE 1

FIGURE 1 shows an ice cube with mass of 150 g and volume of 6.0 cm^3. Calculate the density of the ice cube and state your answer in S.I. unit.

[3 marks]

(b)

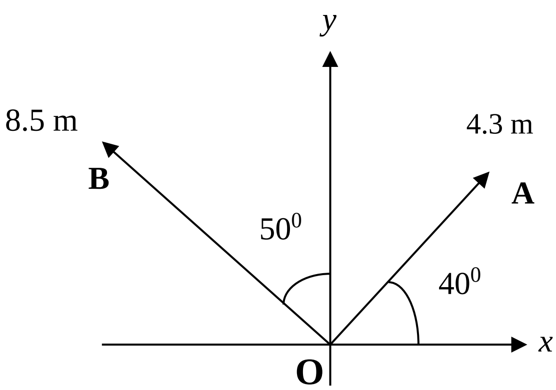

FIGURE 2

FIGURE 2 shows the displacement **OA** and **OB**. Calculate the magnitude and direction of **OA - OB**.

[5 marks]

2. (a)

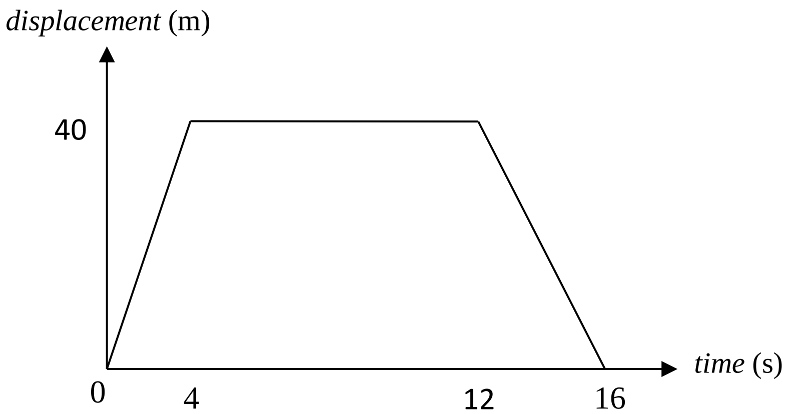

FIGURE 3

FIGURE 3 shows the motion of a body in 16 seconds.

(i) Describe the motion of the body

(ii) Sketch the velocity-time graph.

(iii) Calculate the total distance travelled.

[6 marks]

(b) A boy throws a tomato vertically upward into the air from the ground with an initial velocity of 30 m s^{-1}. Calculate the

(i) maximum height the tomato reaches, and

(ii) time taken to reach the maximum height.

[6 marks]

3. (a) Water melon A of mass 1000 g moving at 6 m s^{-1} collides with another water melon B, of mass 600 g moving at 3 m s^{-1} in the opposite direction. After collision, water melon A bounces backward with velocity 0.5 m s^{-1}. Calculate the

(i) initial momentum of water melon A,

(ii) velocity and state the direction of water melon B after the collision,

[5 marks]

(b)

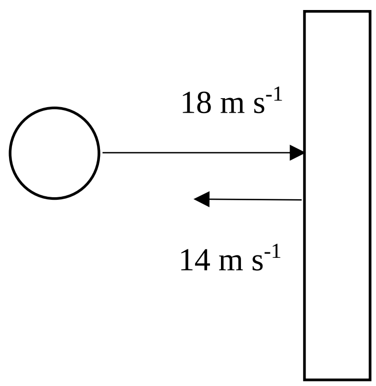

FIGURE 4

FIGURE 4 above shows a sphere of mass 0.50 kg moving with a horizontal velocity of 18 m s^{-1} hits a vertical wall. After 0.06 s, the ball rebounded backward with its final velocity of 14 m s^{-1}. Calculate the

(i) impulse of the sphere

(ii) average force exerted by the sphere on the wall

[4 marks]

4. (a)

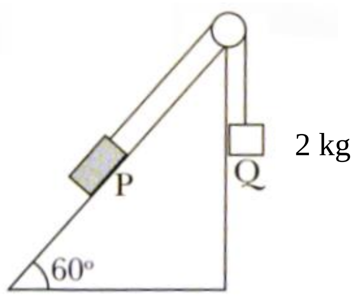

FIGURE 5

FIGURE 5 shows a box, P of mass 5 kg is placed on a rough inclined plank. The box is connected to another block, Q of mass 2 kg by a string through a smooth pulley.

(i) Sketch with label the free body diagrams which show all the forces that act on P and Q respectively.

(ii) When box P is released from rest, it moves under a constant frictional force of 5 N. Calculate the acceleration of box P.

[6 marks]

(b)

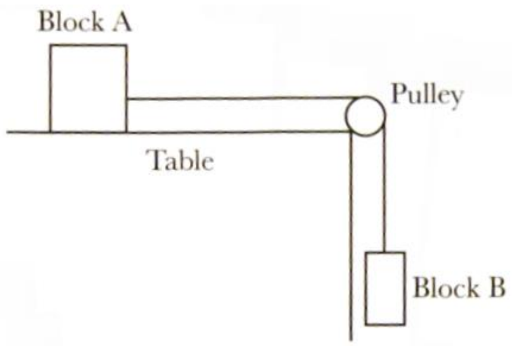

FIGURE 6

FIGURE 6 shows a block A of mass m_A 8.0 kg on a horizontal surface with coefficient of kinetic friction, 0.15 connected by a string through a smooth pulley to another block B of mass m_B 4.0 kg. The system is released from rest. Calculate the

(i) Sketch with label the free body diagrams which show all the forces that act on the block A and B respectively.

(ii) Calculate the acceleration of the system and the tension in the string.

[6 marks]

5. (a) A stone at the end of a string of length 1.2 m is whirled in a horizontal circle on a smooth surface. The speed of the stone is 0.75 m s^{-1}.

(i) Calculate the angular velocity of the stone.

(ii) How many revolutions does the stone make in 2 minutes?

[4 marks]

(b) An object of mass 100 g is placed 50 cm from the centre of a horizontal turn-table. If the maximum frictional force between the object and turn-table is 0.55 N,

(i) draw the free body diagram to show all the forces acting to the object.

(ii) calculate the angular velocity of the turn-table when the object is about to slide.

[5 marks]

6. (a) A rotating wheel of radius 25 cm requires 3 s to rotate 37 revolutions. Its angular velocity at the end of the 3 s interval is 98 rad s^{-1}. Calculate the

(i) initial angular velocity,

(ii) angular acceleration,

(iii) tangential acceleration, and

(iv) angular displacement travelled at the end of the 3 s interval.

[8 marks]

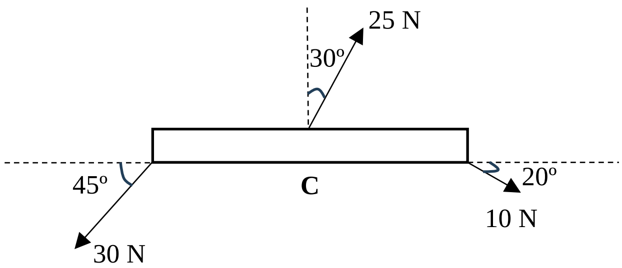

FIGURE 7

(b) Calculate the net torque on the 4 m beam in **FIGURE 7** by taking moment about the centre of the beam, **C**.

[4 marks]

7. (a) An insulated rectangular glass windowpane on a house has a width of 1 m, a height of 2 m, and a thickness of 0.40 cm. On a day when the outside temperature of the house is 36 ºC and the inside temperature of the house is 28 ºC,

(i) calculate the heat transferred through the window by conduction in 12 hours

(ii) sketch the graph of temperature against distance.

(Given coefficient of thermal conductivity of glass is 0.9 W m^{-1} $°C^{-1}$)

[6 marks]

(b) (i) The speed of four molecules in a volume of hydrogen gas are 3.5×10^4 m s^{-1}, 3×10^4 m s^{-1}, 3.6×10^4 m s^{-1} and v_4. Given the root mean square speed of the hydrogen gas is 3.54×10^4 m s^{-1}. Calculate the unknown speed of the fourth molecule, v_4.

(ii) Calculate the temperature of carbon dioxide (molecular mass 44 g mol^{-1}) produced in a fire if the root mean square speed of the gas is 4.72×10^2 m s^{-1}.

(iii) Calculate the percentage of change in the root mean square speed of an ideal gas if the temperature of the gas increase by 40%.

[6 marks]

(c) An ideal gas undergoes two thermodynamics processes as shown in **FIGURE 8**.

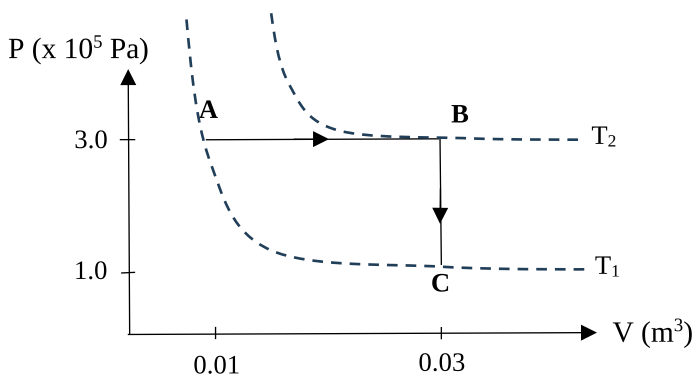

FIGURE 8

(i) Name the processes AB to BC.

(ii) If the initial temperature is 40 ºC, calculate the temperature at state B, T_2.

(iii) Given that work done for process AB is 6000 J. Calculate the work done for the process AC.

(iv) Calculate the total change in internal energy for the process AC.

(v) Calculate the heat needed for the process AC.

[6 marks]

END OF QUESTION PAPER